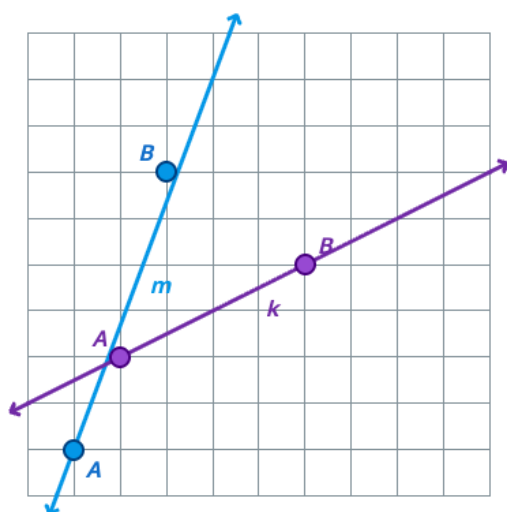


Mini Lesson Sample Script

Materials: Straightedges

1. Compare the steepness of two lines on a grid and find the slope of each using a slope triangle.

Which line is steeper, line m or line k? Then find the slope of each line by drawing a slope triangle between points A and B on each line.



Answer: Line m is steeper; slope of line $m = 6/3 = 2$; slope of line $k = 2/4 = 1/2$.

T: Take a look at these two lines on the grid. If each line were a slide, which would you go down faster on?

S: Line m. It's way steeper.

T: Right, line m is steeper. We can describe how steep a line is with a number called the slope. To find it, we draw a slope triangle between two points and divide the vertical distance by the horizontal distance. Let's try it on line m. Going from A to B, how far do we move across?

S: 3 units to the right.

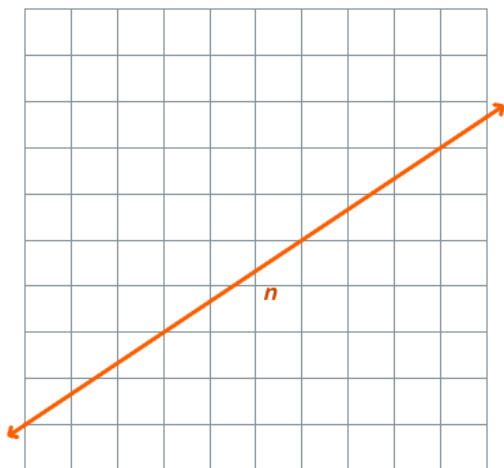
T: And how far up?

S: 6 units up.

T: So the slope of line m is vertical over horizontal, 6 over 3. What's that as a number?

S: 2.

	T: Now try line k. Draw a slope triangle from A to B and tell me the horizontal and vertical distances. S: Across is 4, and up is 2. So the slope is 2 over 4, which is $\frac{1}{2}$. T: Look at the two slopes. Line m has slope 2 and line k has slope $\frac{1}{2}$. Which line did we say was steeper? S: Line m. T: And it has the bigger slope. The bigger the slope, the steeper the line, because the line goes up more for every unit across.
2. Find the slope of a line on a grid by drawing a slope triangle between two points.	
Find the slope of line n. Choose two points where the line crosses the intersection of grid lines, then draw a slope triangle to help you.	T: Pick two points where line n crosses the intersection of grid lines. Why is that a smart choice? S: Because then I can count whole units. If I pick somewhere in the middle, I'd have to guess. T: Good. Draw a slope triangle between your two points and find the slope.



Answer: Slope of line $n = 2/3$ (e.g., from $(0, 1)$ to $(3, 3)$, vertical distance 2 and horizontal distance 3).

S: (Draws.) I went 3 across and 2 up. So the slope is $2/3$.

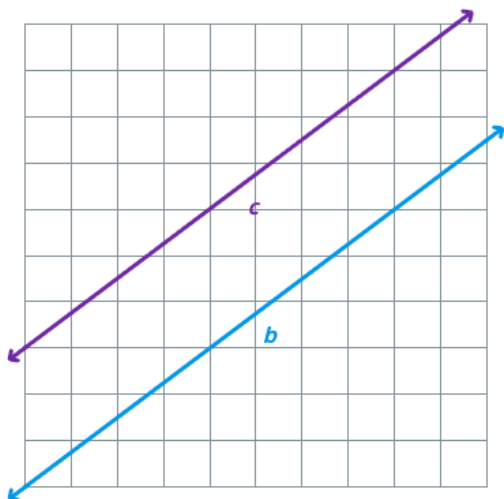
T: What if you picked a different pair of points on line n ? Would you get the same slope?

S: I think so, because the line doesn't change.

T: That's right. Any slope triangle you draw on the same line will be similar to the others, so the ratio of vertical to horizontal stays the same.

3. Find the slope of a line parallel to a given line by drawing and comparing slope triangles.

Lines b and c are parallel. The slope of line b is $3/4$. Find the slope of line c by drawing a slope triangle on line c . What do you notice?



T: The slope of line b is $3/4$. That means for every 4 units across, the line goes up 3. Now look at line c . Pick two lattice points on line c and draw a slope triangle.

S: (Draws from $(0, 3)$ to $(8, 9)$.) Across is 8, up is 6.

T: So what's the slope of line c ?

S: 6 over 8, which is $3/4$.

T: Compare that to the slope of line b . What do you notice?

S: They're the same. Both are $3/4$.

Answer: Slope of line $c = 6/8 = 3/4$. The slopes are the same; parallel lines have equal slopes.

T: That makes sense, right? Parallel lines have the same steepness, so they have the same slope. Their slope triangles have the same shape, which means the ratio of vertical to horizontal is the same.